

Chapter 9: Carbonyl Compounds

PSPM 2011 / 2012

- Q. 1 (a) Treatment of 1-bromobutane with sodium cyanide gives *I*. Hydrolysis of *I* forms *J*. 3 m
- (i) Draw the structures of *I* and *J*.
- (ii) Give IUPAC name of compound *J*

Answer (a)(i) $I = \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CN}$
 $J = \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$

(ii) Pentanoic acid

- Q. 1 (b) Suggest the products formed when *J* reacts with 4 m
- (i) methanamine with heating.
- (ii) LiAlH_4 followed by hydrolysis.
- (iii) SOCl_2 .
- (iv) ethanoic acid.

Answer

(b)(i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \overset{\text{H}}{\text{N}} - \text{CH}_3$

(b)(ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(b)(iii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{Cl}$

(b)(iv) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{O} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$

- Q. 1 (c) Indicate the reagents needed for the following conversions: 3 m
- (i) butanone to 2-butanol.
- (ii) butanone to 2-hydroxy-2-cyanobutane.
- (iii) benzene to acetophenone

Answer (c)(i) NaBH_4 or LiAlH_4 followed by H_3O^+ @
 or
 H_2 , Ni / Pd / Pt

(c)(ii) HCN or KCN or NaCN or CN^-

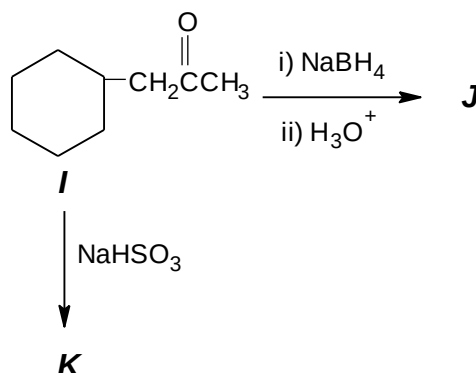
(c)(iii) CH_3COCl , AlCl_3 or FeCl_3
 @ CH_3COBr , AlBr_3 or FeBr_3

(i) LiAlH_4
 $\xrightarrow{\hspace{1cm}}$
 (ii) H_3O^+

PSPM 2012 / 2013

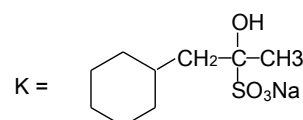
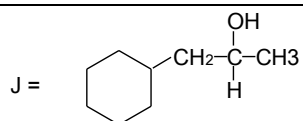
Q. 2 (a) Reactions of compound *I* are shown in the following scheme.

6 m



- (i) Draw the structure for *J* and *K*.
- (ii) What is the nucleophile involved in the formation of *J*?
- (iii) Suggest a chemical test to differentiate between *I* and *J*. State the observation and write the chemical equation involved.
- (iv) Show how *I* can be synthesized from an alkyl halide containing nine carbon atoms.

Answer (a)(i)

 H^- (hydride ion)

- (ii) Lucas test ($\text{ZnCl}_2, \text{concHCl}$) OR Brady's test (2,4-DNPH)
 Lucas test : *J* gives cloudy solution within 5 minutes
I no observable change

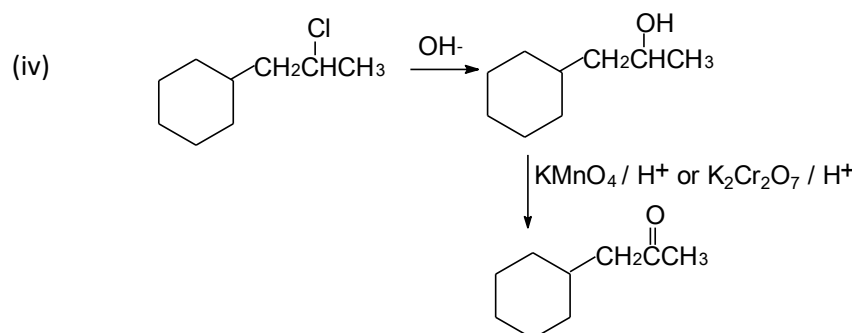
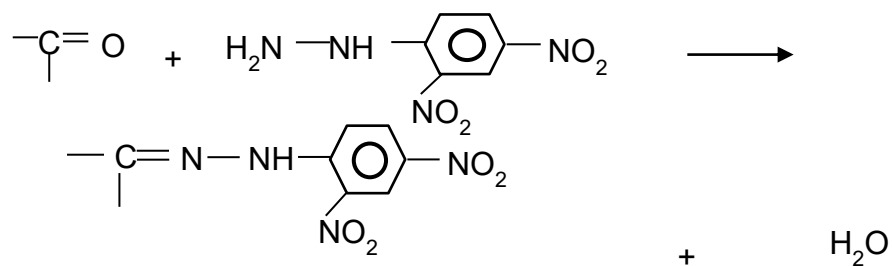
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- (iii) Equation:
 $\text{C}_6\text{H}_{11}\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 + \text{ZnCl}_2 / \text{HCl (Conc)} \longrightarrow$
 $\text{C}_6\text{H}_{11}\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$

OR

Brady's test : *I* gives brick red / orange precipitate / yellowish orange
J no observable change

}



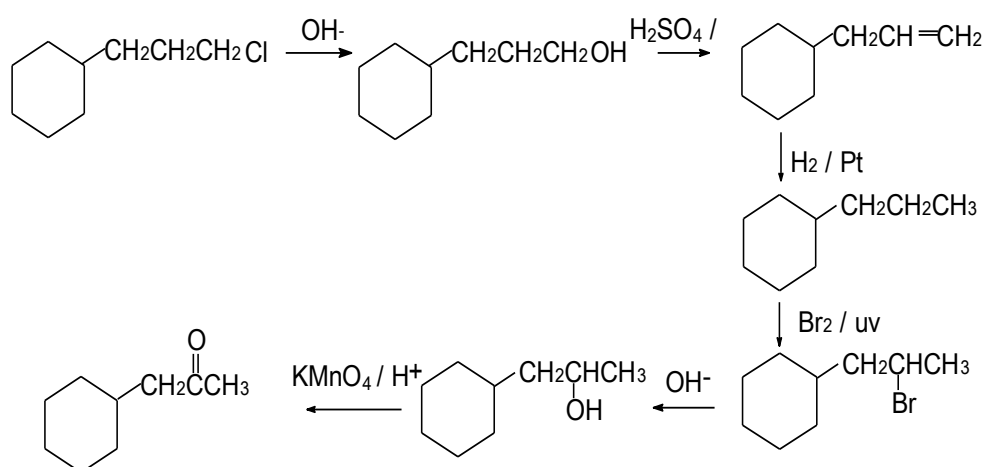
Note :

1M – Starting material

2M – Reagents

1M- Intermediate

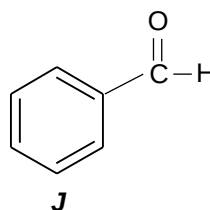
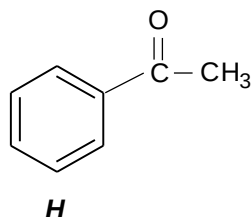
OR



PSPM 2013 / 2014

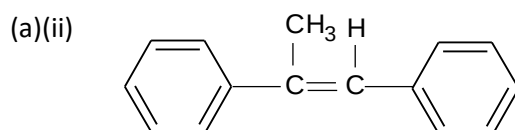
Q. 3 (a) Ozonolysis of compound **G** gives compounds **H** and **J**.

10 m



- (i) State the reagents used in the ozonolysis reaction.
By using a suitable explanation, propose the structure of **G**.
- (ii) Draw the structures of the intermediates and final products formed when **H** and **J** react with $(\text{CH}_3)_2\text{CHMgBr}$, followed by hydrolysis.
Explain why **H** reacts with an alkaline solution of iodine whereas **J** does not.
Write the chemical equation for the reaction.

Answer (a)(i) ozone (O_3) ii. $(\text{CH}_3)_2\text{S}$ or ozone (O_3) ii. $\text{Zn}, \text{H}_2\text{O}$

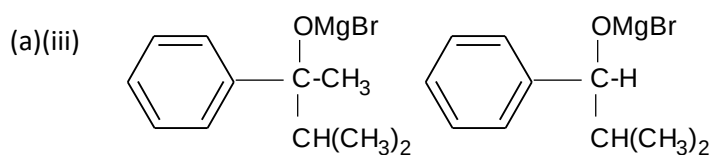


Ozone **cleaves the C=C** of alkene to give carbonyl compounds

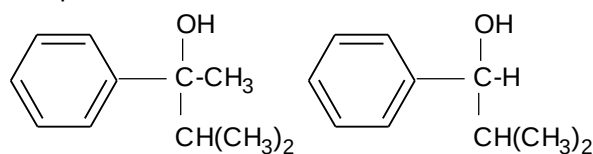
Or

Ozonolysis is used to **confirm the position of C=C**

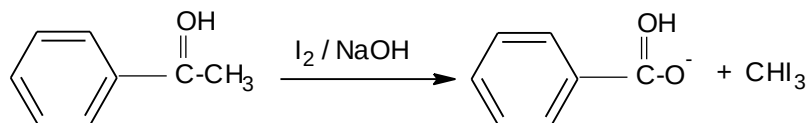
Intermediates:



Final products:



(a)(iv) **H** has a methyl ketone group, $\text{CH}_3\text{C}=\text{O}$



PSPM 2014 / 2015

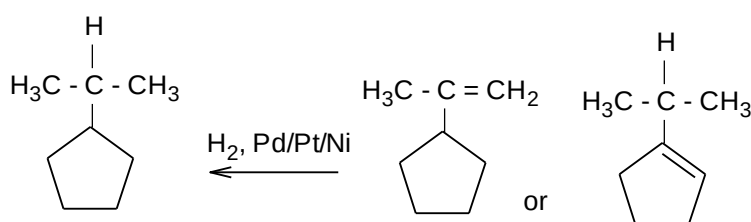
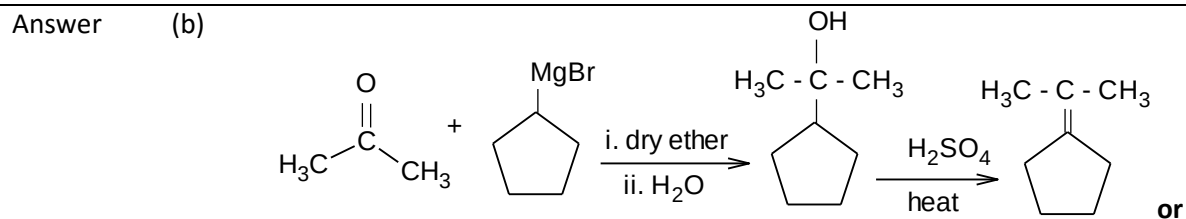
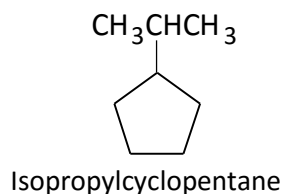
- Q. 4 (a) Suggest suitable reagents for the conversion of butanal to the following compounds: 2 m
- (i) 1-Butanol
- (ii) Butanoic acid

Answer (a)(i) LiAlH_4

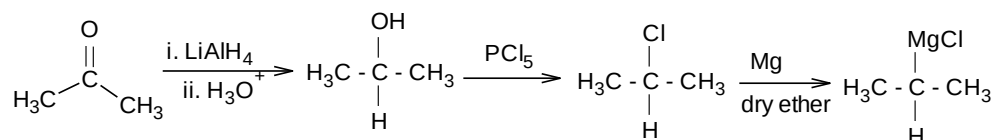
(a)(ii) H_3O^+ @ $\text{NaBH}_4, \text{CH}_3\text{OH}$ @ H_2 / cat

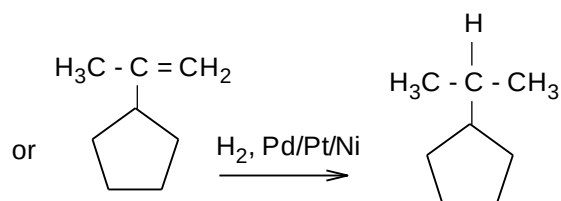
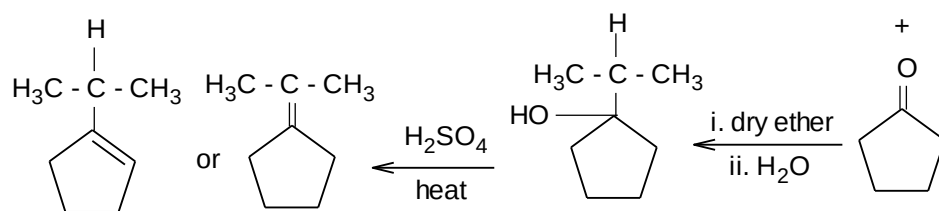
cat = Pd, Ni, Pt, Pd-C

- Q. 4 (b) Show the synthetic steps for the preparation of isopropylcyclopentane from propanone via Grignard reagent. 5 m

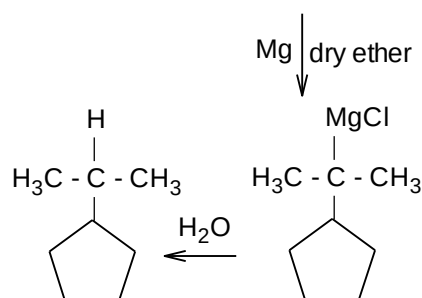
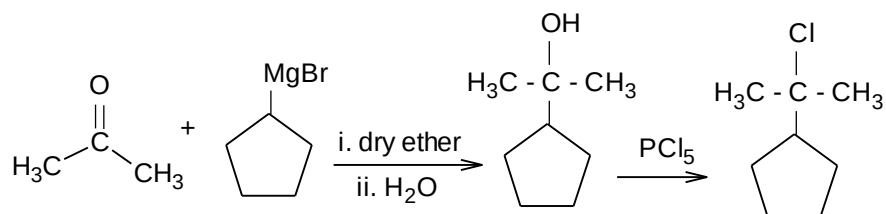


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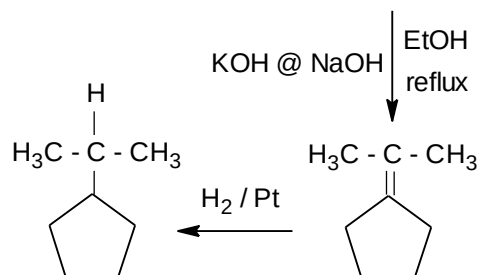
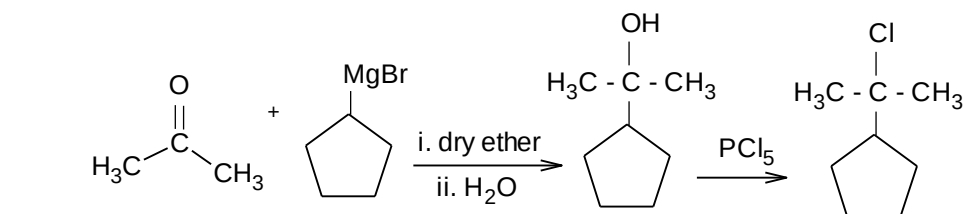




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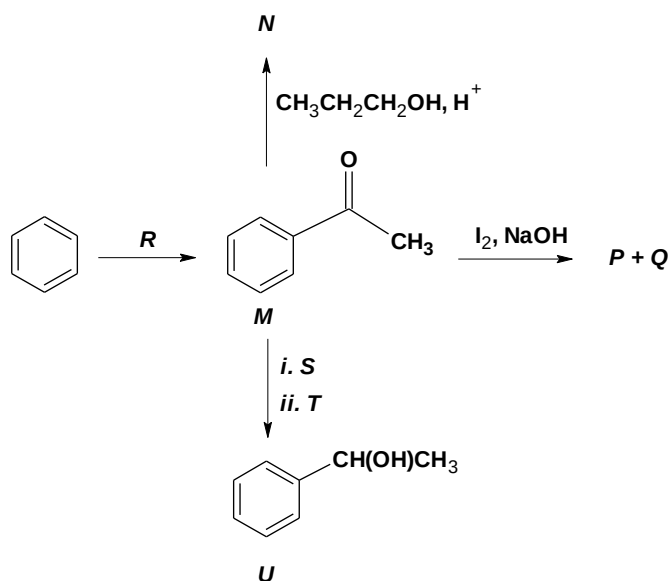
OR



PSPM 2015 / 2016

Q. 5 (a) Reactions of acetophenone, M, is shown below:

9 m

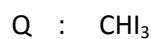
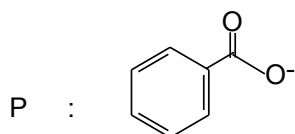
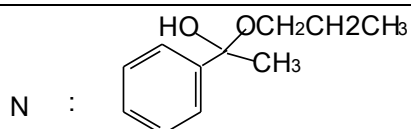


- (i) Draw the structures of N, P and Q.
- (ii) Give the reagents of R, S and T.
- (iii) Suggest a chemical test to differentiate M from U. State the observation and write the reaction equation.

Answer

(a)

(i)



(ii)

R : CH_3COCl , AlCl_3 @ other lewis acid catalyst

S : LiAlH_4 @ NaBH_4

T : H_3O^+ @ CH_3OH

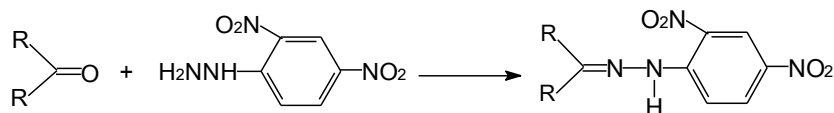
(iii)

Brady's test

Observation **M** : gives yellow / orange precipitate

U : No yellow / orange precipitate form
@ no observable change.

}



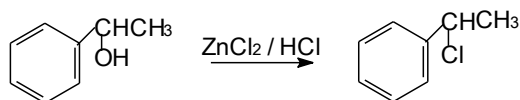
OR

Lucas test

Observation **M** : no observable change.

U : Cloudy solution formed

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PSPM 2017 / 2018

Q. 6 (a) 3-methyl-2-phenyl-2-butanol can be prepared from the reaction between carbonyl compounds and Grignard reagents.

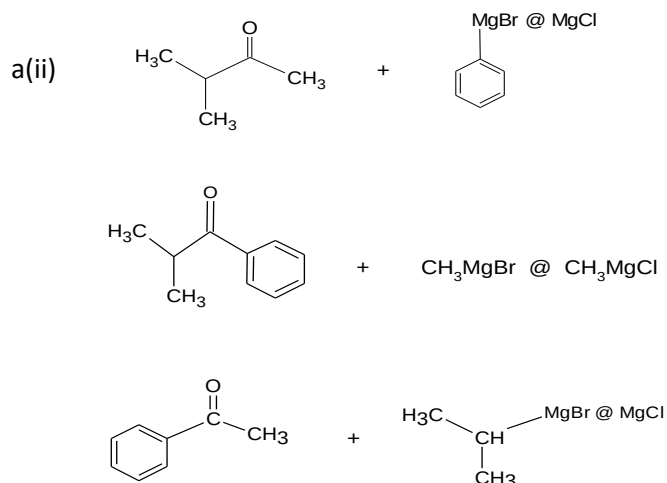
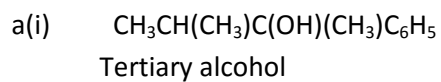
(i) Draw the structure and state the class of this alcohol.

2 m

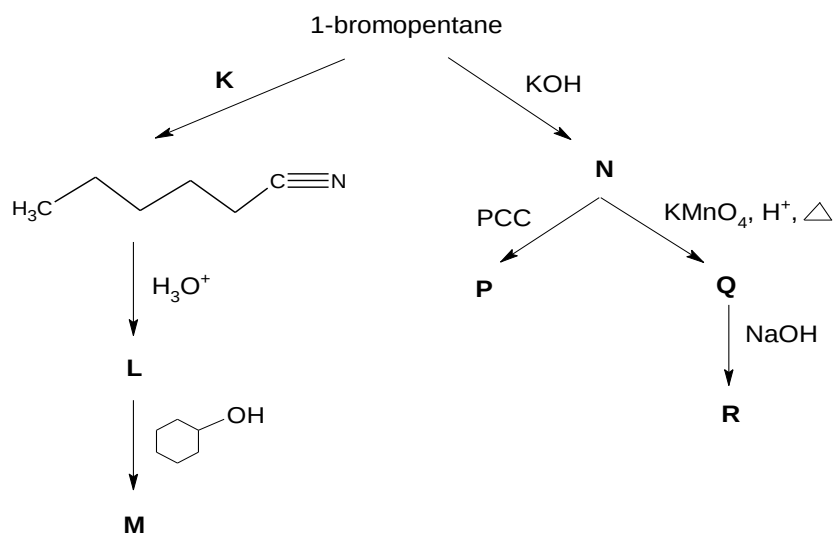
(ii) Draw all the possible pairs of carbonyl compounds and Grogard reagents that can be used to form 3-methyl-2-phenyl-2-butanol.

6 m

Answer



Q. 6 (b) The reaction of 1-bromopentane is shown in the scheme below.



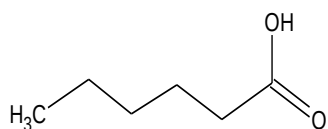
7 m

- (i) Draw the structures of **L**, **M**, **N**, **P**, **Q** and **R**.
(ii) State reagent **K**.

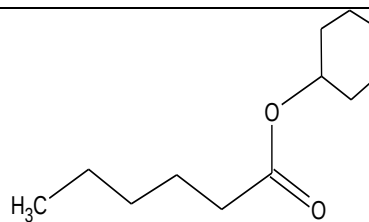
Answer

(b)

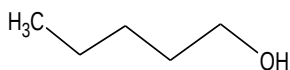
(i)



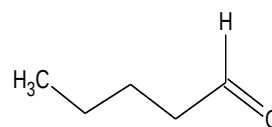
L



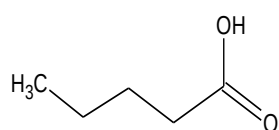
M



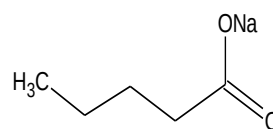
N



P



Q



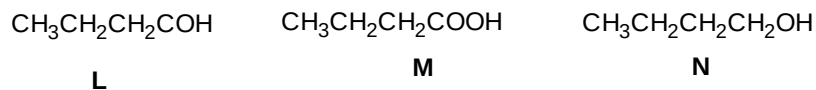
R

(ii)

K : KCN @ NaCN @ CN^-

PSPM 2018 / 2019

- Q. 7 (a) Arrange the following compound in ascending order of boiling point. Explain your answer. 8m



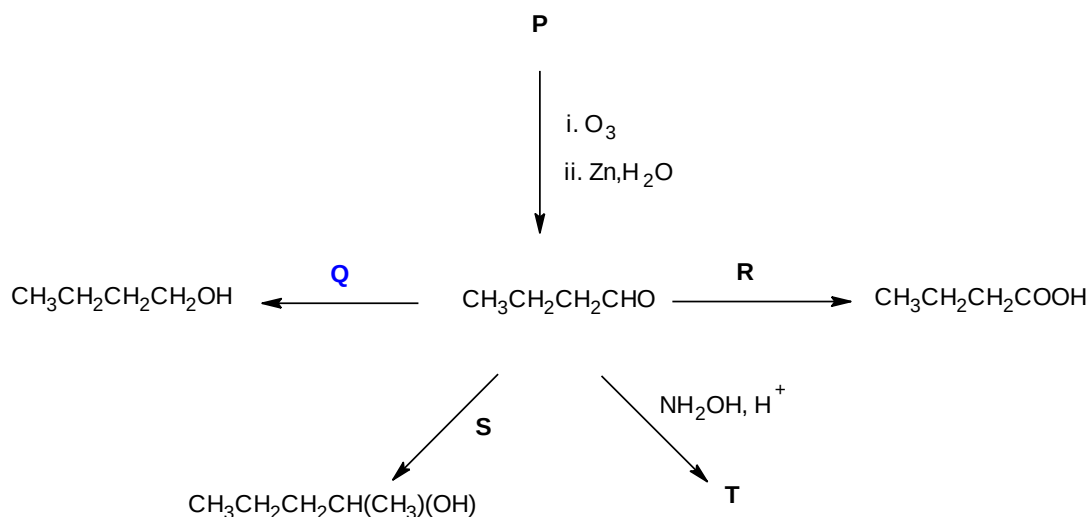
Answer Arrangement : $\text{L} < \text{N} < \text{M}$

Explanations :

M has the highest boiling point because of dimerization/exists as dimers.

N has higher boiling point than L because the hydrogen bonding in N is stronger than dipole-dipole interactions in L.

- Q.8 (b) Based on the following reaction scheme, draw the structural formulae of P and T, and give reagents Q, R and S.



Answer

P : $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_2\text{CH}_3$

Q : LiAlH_4 followed by H_3O^+ @ (i) LiAlH_4 (ii) H_3O^+
 @ $\text{NaBH}_4, \text{CH}_3\text{OH}$
 @ $\text{H}_2, \text{Ni/Pt/Pd}$

R : $\text{KMnO}_4, \text{H}^+, \text{heat}$ @ $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+, \text{heat}$ @ $\text{CrO}_3, \text{H}^+, \text{heat}$

S : i) CH_3MgBr ii) H_3O^+

T : $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{NOH}$